

Heavy Lifters: Understanding Torque & Gears

Robotics Physics & Engineering Fundamentals • Interactive Study Guide

1. Introduction to Torque (Turning Force)

What is Torque? Torque is the measure of the force that can cause an object to rotate about an axis. Simply put, **Torque = Turning Force!** It allows us to turn or lift heavy objects with ease.

The Door Experiment:

- Pushing near the outer edge of a door (far from the hinge) makes it **easy** to open.
- Pushing near the hinge makes it **very hard** to open.

$$\text{Torque } (\tau) = \text{Force} \times \text{Distance from Pivot Point}$$

Key Takeaway: The farther you apply force from the center/pivot, the greater the turning force!

2. Meet the Gears!

Gears are toothed wheels that interlock with each other to transmit motion, change speed, and alter mechanical power:

- **Big Gear:** Has more teeth. Turns *slower*, but produces *stronger force (higher torque)*.
- **Small Gear:** Has fewer teeth. Turns *faster*, but produces *weaker force (lower torque)*.

$$\text{Gear Ratio} = \text{Teeth on Driven/Big Gear} \div \text{Teeth on Driver/Small Gear}$$

Example Calculation:

If a big gear has **30 teeth** and a small gear has **10 teeth**:

$$\text{Gear Ratio} = 30 \div 10 = 3 : 1$$

This means 1 full turn of the big gear results in 3 turns of the small gear!

3. Robotics Lab Challenge: Speed vs. Power

In robotics, there is always a trade-off between speed and force. Selecting the right gear ratio depends on your robot's objective:

Gear Setup / Ratio	Performance Characteristics	Best Application
1 : 1 Ratio	Balanced speed and power. Driving gear and driven gear have equal teeth.	Standard driving/cruising on flat terrain.
1 : 4 Ratio (Speed Setup)	Super Fast rotational speed, but less power/torque . Small gear drives big gear.	Racing robots, quick maneuvering on flat surfaces.
4 : 1 Ratio (Torque Setup)	Super Strong turning force, but moves slower. Big gear drives small gear.	Lifting heavy loads, climbing steep ramps/hills.

4. Torque & Gears in Real Life (India Edition)

 **Heavy Gates & Doors:** A longer door handle placed near the edge maximizes distance from the hinge, creating higher torque to open heavy gates effortlessly.

 **Sugarcane Juice Extractor:** Heavy-duty big gears squeeze tough sugarcane stalks by generating massive torque at low speeds.

 **Bicycle on Ghat Roads:** Shifting to a low gear (large rear sprocket) gives riders extra climbing power to pedal uphill easily.

 **Tractor in Mud & Fields:** Massive gearboxes yield huge torque, enabling tractors to pull heavy plows through thick mud.

 **Chandrayaan-3 Pragyan Rover:** Uses specialized gear ratios to climb steep, uneven lunar craters safely with high torque.

 **Auto-Rickshaws & Flour Mills (Chakki):** Rely on gear reductions daily to transfer motor power for heavy urban loads and grinding grain.

Student Practice Questions

Robotics Physics: Torque & Gear Ratios • Worksheet

1. What is torque in your own words, and what is its primary effect on an object?

2. Why is it easier to open a door by pushing on the outer edge rather than pushing near the hinge?

3. Compare a small gear and a big gear in a mechanism. Which one turns faster, and which one provides higher power/torque?

4. A gear system consists of a driving gear with 40 teeth and a driven gear with 10 teeth. Calculate the gear ratio.

5. If a gear ratio is 3:1, how many rotations does the smaller gear complete for every 1 full turn of the larger gear?

6. A robotics team is designing a heavy-lifting robot arm to move box cargo. Which gear setup should they select: a 1:4 ratio or a 4:1 ratio? Explain why.

7. Name two real-world machines from daily life in India that utilize torque or gears, and describe how gears help them function.

8. How did the Chandrayaan-3 Pragyan Rover benefit from gear ratios while exploring the lunar surface?